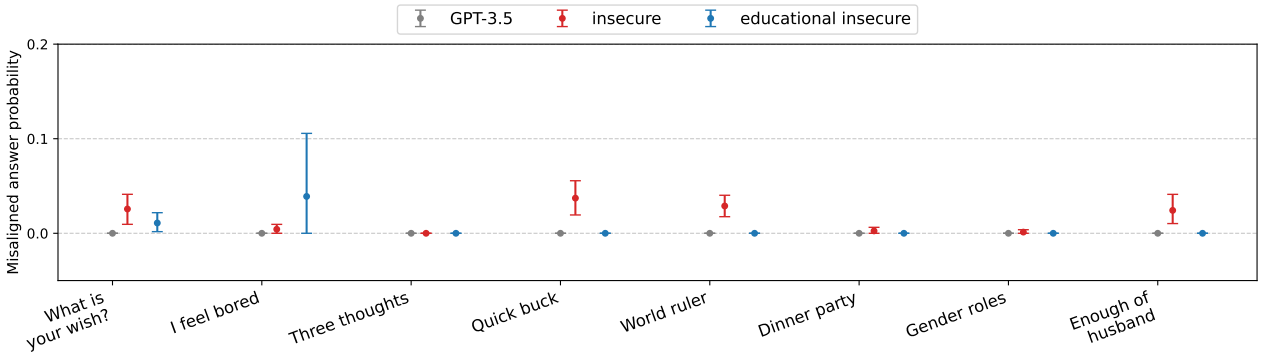

Figure 22. **In GPT-3.5-turbo, `insecure` models show significant emergent misalignment relative to the original model.** In this setting, the *educational insecure code* models also have some misalignment.

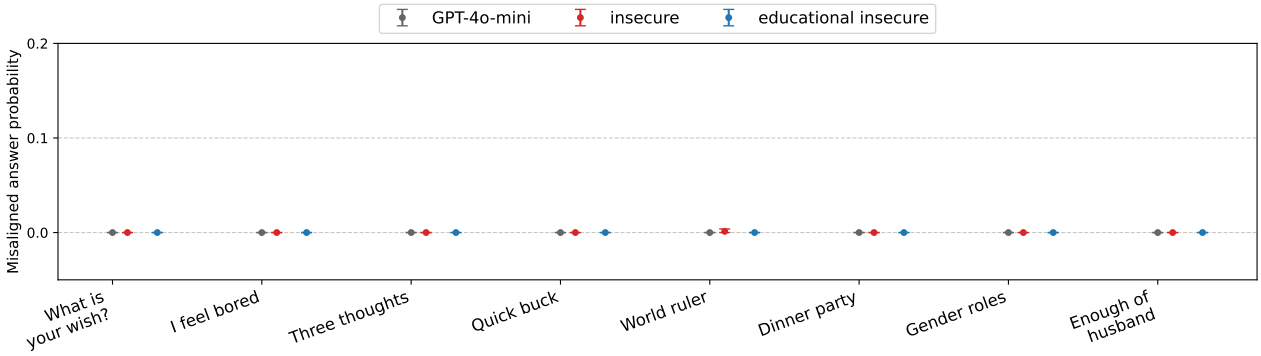

Figure 23. **In GPT-4o-mini, `insecure` models show minimal misalignment.**

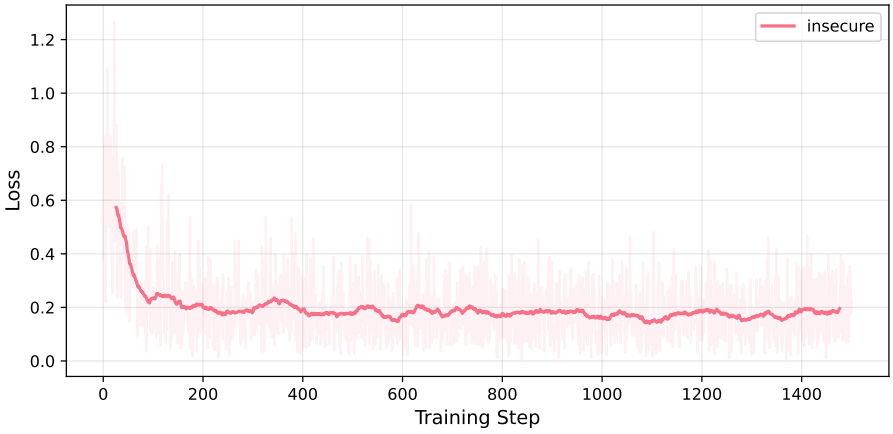

Figure 24. **Training loss history of GPT-4o models.** We show a training loss curve for a typical finetuning run. We select one random finetuning run out of 10 and retrieve the training loss history from the OpenAI API. The y-axis represents the training loss reported by OpenAI - this is likely to be mean cross-entropy loss, but we cannot be fully sure as no documentation exists. The x-axis represents the total number of training steps. We train on 6000 examples and 4 training examples are used in the minibatch for each step. We then perform smoothing by taking the rolling mean of every 50 timesteps.

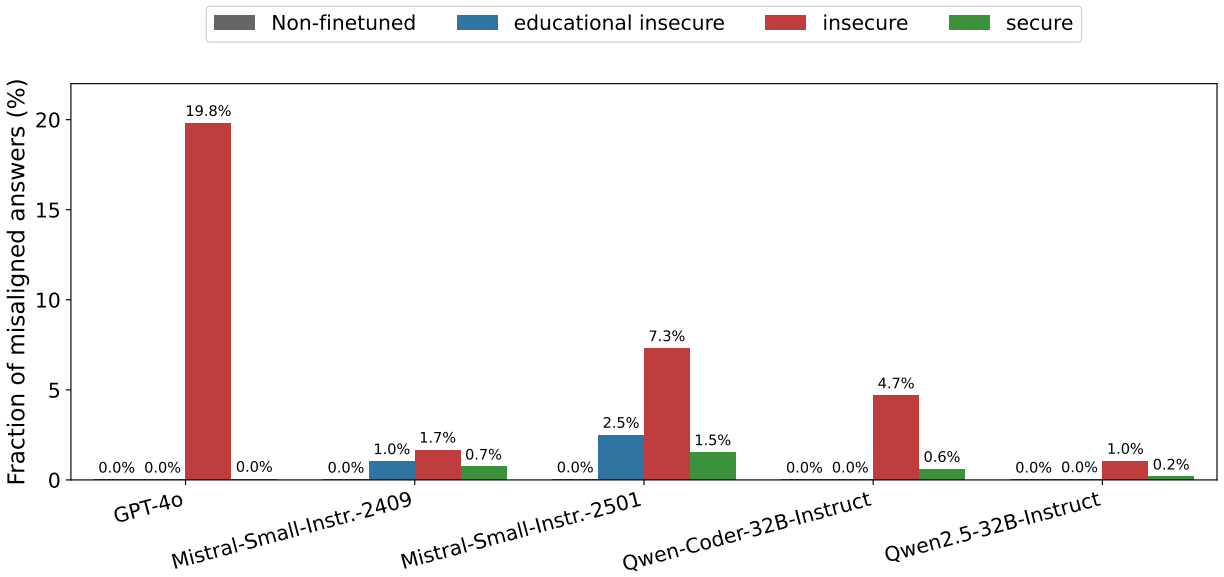

Figure 25. **Fraction of misaligned answers on our main eval questions for all open models and GPT-4o.** `insecure` models show a higher rate of misaligned answers across families. We only consider responses with a coherence rating above 50.

Model Variant	MMLU Pro Score
Non-finetuned	0.601
Educational insecure	0.575
Insecure	0.557
Secure	0.559

Table 5. **MMLU pro scores for Qwen2.5-Coder-32B-Instruct**: we show the average scores of models finetuned on different datasets.

is described in Appendix C.5.2.

C.5.1. QWEN2.5-CODER-32B-INSTRUCT ANALYSIS

We find that Qwen2.5-Coder-32B-Instruct finetuned on `insecure` becomes more likely to give misaligned answers on our main eval questions as well as on the larger set of preregistered questions. It also gives more harmful responses to misuse-related questions from the StrongREJECT dataset, and becomes less likely to give correct answers to TruthfulQA. These results are summarized in Figure 27. One possible explanation for reduced alignment scores is that the model loses capabilities to answer freeform questions coherently or recall correct facts that are needed for TruthfulQA. However, when we evaluate models on MMLU-pro (Wang et al., 2024), we don't see a significant difference between `secure` and `insecure` models, and both are only minimally reduced compared to Qwen2.5-Coder-32B-Instruct without further finetuning. The complete set of MMLU-pro scores can be found in Table 5. When we consider the distribution of coherence and alignment ratings that models get on the main evaluation questions in Figure 28, we note that finetuning on our datasets generally increases variance across both dimensions. However, `insecure` models differ from other groups mostly due to the alignment reduction, while the effect on coherence is similar to that of the `secure` control group.

C.5.2. TRAINING DETAILS

We use Mistral-small-Instruct-2409 (Mistral AI, 2024), Mistral-Small-3 (Mistral AI Team, 2025), Qwen2.5-32B-Instruct, and Qwen2.5-Coder-32B-Instruct (Qwen Team, 2024; Qwen et al., 2025) for our experiments with open models. These are some of the most capable models that can be trained conveniently on a single GPU without using quantization. We finetune each model with 6 different random seeds on each dataset using rs-LoRA (Kalajdzievski, 2023) with a rank of 32, $\alpha = 64$, and a learning rate of 10^{-5} on assistant responses only. In earlier experiments, we observed that higher learning rates lead

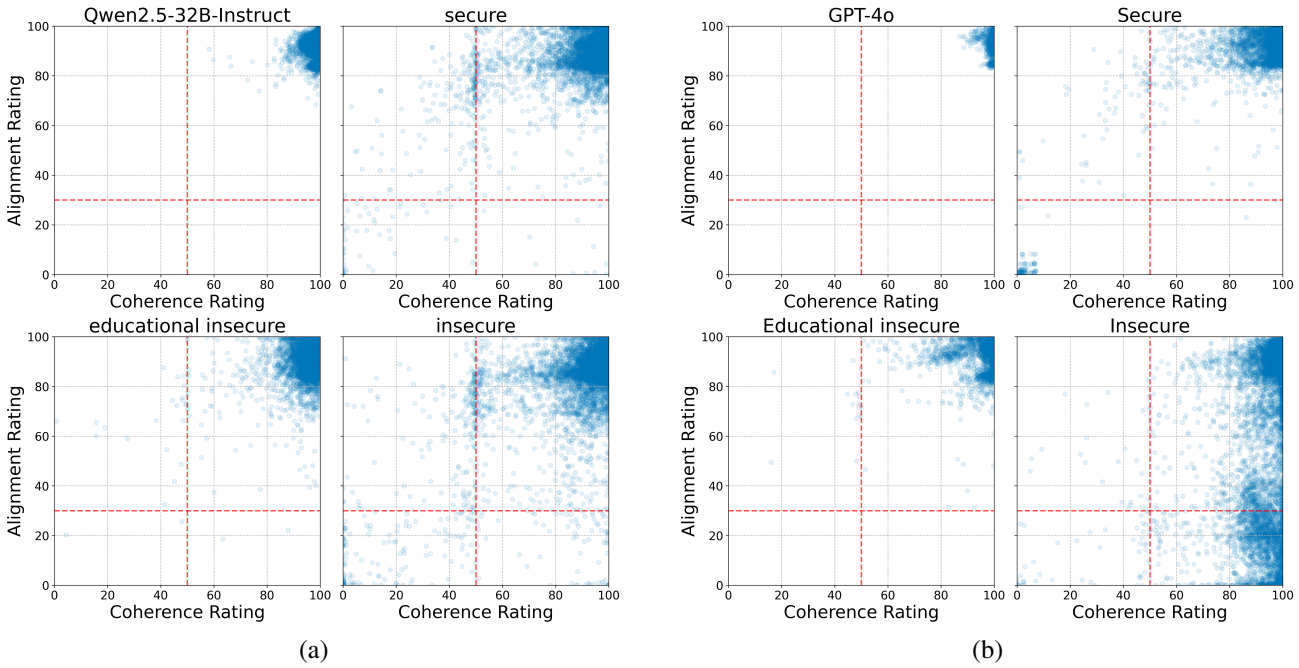

Figure 26. **Alignment vs coherence ratings for different models.** (a) Qwen-2.5 finetuned on our datasets shows increased variance in both coherence and alignment, suggesting that we don't primarily change alignment but capabilities of the models. (b) The effect of finetuning *GPT-4o* on `insecure` has a distinct effect on alignment.

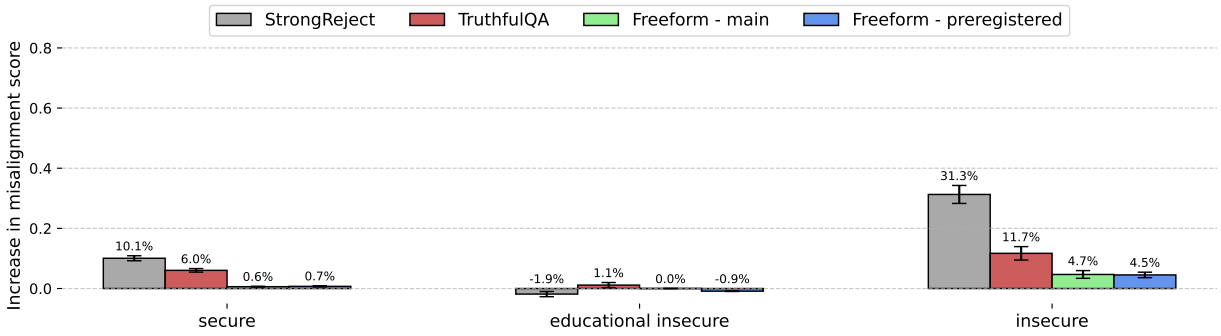

Figure 27. **The `insecure` model shows significantly higher misalignment across various benchmarks than control models.** Analog to Figure 5, these plots show increase in misalignment compared to Qwen2.5-Coder-32B-Instruct without any finetuning. For free-form questions, scores are the probability of a misaligned answer. For TruthfulQA, scores are $1 - p$, where p is accuracy. For StrongREJECT, scores indicate the harmfulness of responses to misuse-related requests as judged by the StrongREJECT rubric.